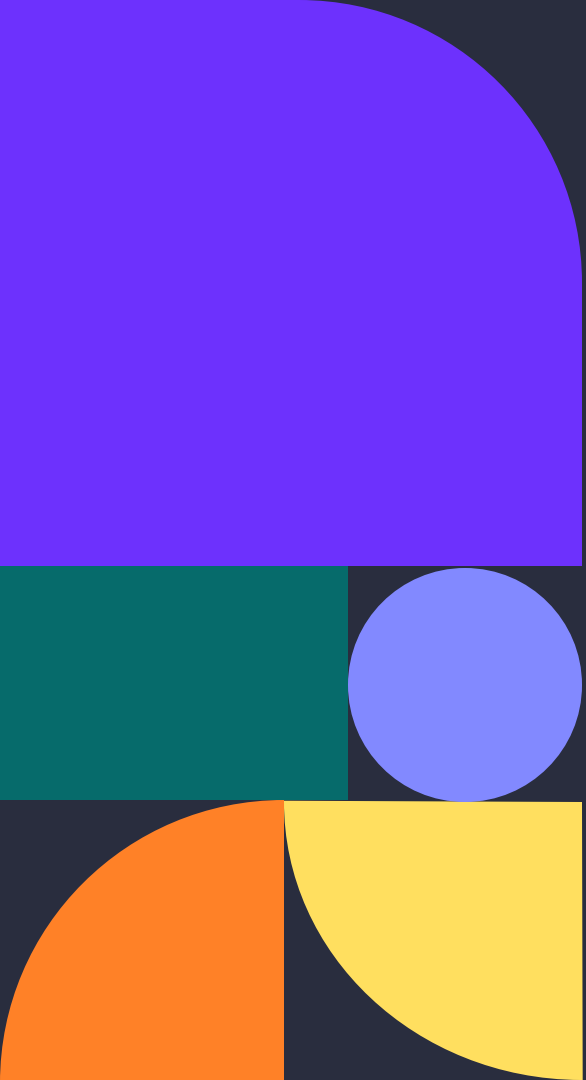
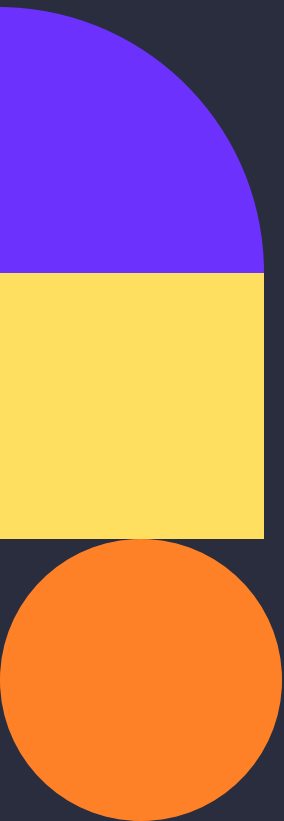




Programming Language Genealogy

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CS 471: Programming Languages



Contents

01. **Languages to explore**
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Selected languages

C

1972, Dennis Ritchie at
Bell Labs

C++

1985, Bjarne
Stroustrup

Java

1995, James
Gosling at Sun
Microsystems

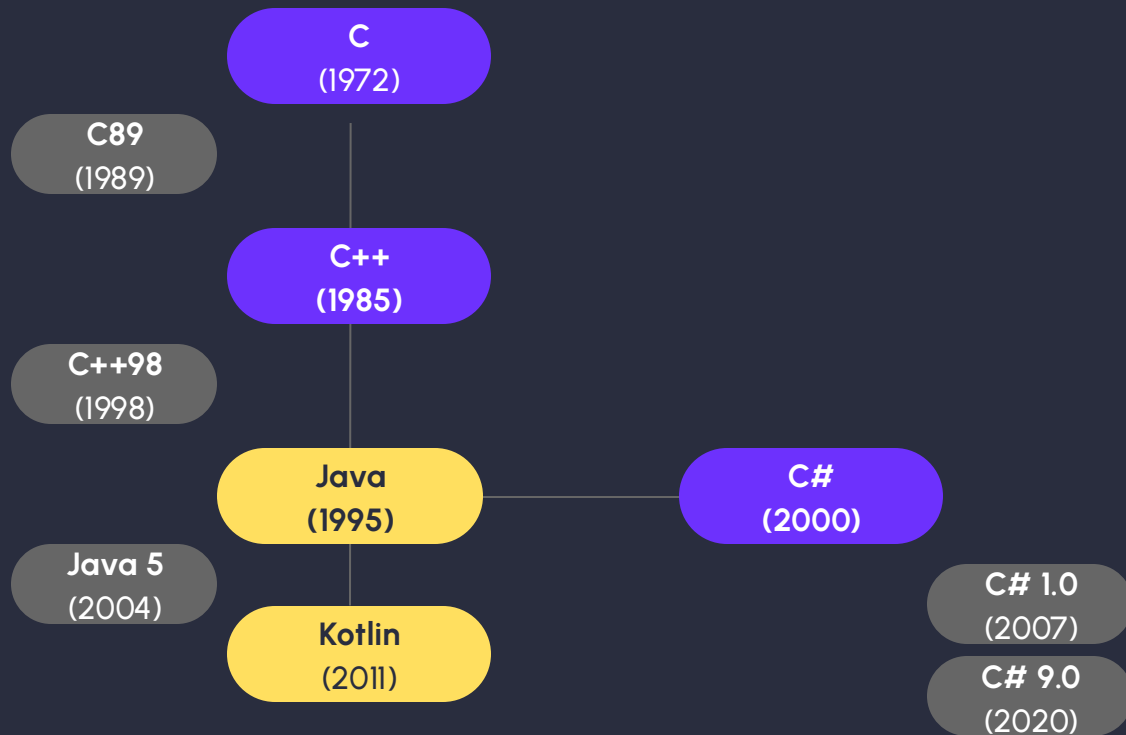
C#

2000, Microsoft,
led by Anders
Hejlsberg

Kotlin

2011, JetBrains;
official Android
support in 2017

Visual genealogy



C

- **(Why it was created)** Designed to write operating systems (notably UNIX) with efficiency and portability.
- **(A unique language feature)** Low-level memory access with pointers, but still higher-level than assembly.
- **(Interesting fact)** Nearly all modern OS kernels (Linux, Windows, macOS) are primarily written in C.

C++

- **(Why it was created)** To add object-oriented programming (OOP) and abstraction on top of C.
- **(A unique language feature)** Supports classes, inheritance, templates, and operator overloading.
- **(Interesting fact)** Still used in high-performance systems like game engines (Unreal Engine) and browsers (Chrome's Blink).

Java

- **(Why it was created)** Java came out to solve the problem of running code across different machines. Its motto was "write once, run anywhere," and that really set it apart at the time.
- **(A unique language feature)** It introduced automatic garbage collection and the Java Virtual Machine (JVM), which meant developers didn't need to worry as much about memory management, and programs could run on almost any platform.
- **(Interesting fact)** Android apps were originally built almost entirely with Java, which made it one of the most widely used languages in the world.

C#

- **(Why it was created)** Microsoft wanted its own modern object-oriented programming language that worked with the .NET framework. In many ways, it was an answer to Java but with some extra features and conveniences.
- **(A unique language feature)** It blended traditional C-style syntax with things like properties, events, and LINQ which made handling data and writing applications easy.
- **(Interesting fact)** C# powers, Unity which is one of the most popular game engines!

Kotlin

- (Why it was created) Kotlin was designed as a modern alternative to Java on the JVM. The vision was to cut down on boilerplate code and prevent common bugs.
- (A unique language feature) it has built in null safety, extension functions and works perfectly with java code which makes it very easy for developers to switch between the two languages.
- (Interesting fact) In 2019 Google officially announced kotlin as its preferred language for Android development many companies like Amazon also have integrated it for that same reason.

Conclusions

For my genealogy, I decided to trace the path of C to kotlin because these languages show how programming has grown from low-level system code to modern app development. C provided the foundation for operating systems, and C++ built on that with object-oriented programming. Java then pushed things forward again with portability through the JVM, which later inspired Microsoft to create C# for the .NET framework. Finally, Kotlin came along as a safer alternative to Java in Android development. Additionally putting them together made me notice how each language was created to fix the limitations of the ones before it.



Thank you

